

Crucible Specifications

ForgeCore Industries — Engineering Division

Engineering Specifications — FC Series Crucibles

This document provides detailed engineering specifications for all standard crucibles in the ForgeCore FC series. All dimensions are nominal and subject to $\pm 2\%$ manufacturing tolerance. Custom specifications are available on request.

FC-45 — Silicon Carbide

Physical Properties

Bulk Density	3.05 g/cm ³
Apparent Porosity	12% max
Cold Crushing Strength	120 MPa min
Modulus of Rupture	45 MPa @ 25°C
Thermal Conductivity	85 W/m·K @ 1000°C
Thermal Expansion	4.5×10^{-6} /°C

Operating Limits

Maximum Service Temperature	1,500°C
Recommended Operating Range	1,100-1,450°C
Thermal Shock Resistance	ΔT 400°C
Maximum Heating Rate	200°C/hour

Standard Dimensions (mm)

Size	Height	Outer Ø	Wall	Capacity
S	200	150	12	2.5 L
M	280	200	15	6.0 L
L	360	260	20	14 L
XL	450	320	25	26 L

FC-50 — High-Purity Graphite

Physical Properties

Bulk Density	1.78 g/cm³
Carbon Content	99.5% min
Ash Content	0.3% max
Electrical Resistivity	12 μΩ·m max
Thermal Conductivity	110 W/m·K
Modulus of Elasticity	10 GPa

Operating Limits

Maximum Service Temperature	1,800°C
Recommended Operating Range	1,200-1,750°C
Oxidation Resistance	Requires inert atmosphere above 600°C
Maximum Heating Rate	300°C/hour

Standard Dimensions (mm)

Size	Height	Outer Ø	Wall	Capacity
S	220	160	10	3.0 L
M	300	220	13	7.5 L
L	400	290	18	19 L
XL	500	360	22	36 L
XXL	600	440	28	65 L

FC-72 — Clay-Graphite Composite

Physical Properties

Bulk Density	2.10 g/cm ³
Carbon Content	32%
Modulus of Rupture	18 MPa
Thermal Conductivity	25 W/m·K
Thermal Expansion	3.2 × 10 ⁻⁶ /°C

Operating Limits

Maximum Service Temperature	1,300°C
Recommended Operating Range	700-1,250°C
Thermal Shock Resistance	ΔT 350°C
Maximum Heating Rate	150°C/hour

Standard Dimensions (mm)

Size	Height	Outer Ø	Wall	Capacity
S	180	130	10	1.8 L
M	260	190	13	5.0 L
L	340	250	17	12 L

FC-90 — Zircon

Physical Properties

Bulk Density	3.85 g/cm ³
ZrO ₂ Content	65% min
SiO ₂ Content	33%
Cold Crushing Strength	150 MPa
Thermal Expansion	4.0 × 10 ⁻⁶ /°C
Thermal Conductivity	5 W/m·K

Operating Limits

Maximum Service Temperature	2,000°C
Recommended Operating Range	1,400-1,950°C
Thermal Shock Resistance	ΔT 250°C
Maximum Heating Rate	120°C/hour

Standard Dimensions (mm)

Size	Height	Outer Ø	Wall	Capacity
S	180	130	10	1.5 L
M	260	190	14	4.5 L
L	360	260	20	11 L

Testing and Certification

Each production lot is sampled and tested according to ASTM C401, C417, and C133 standard test methods. Certificates of analysis (COA) ship with every order. Lot traceability is maintained for 7 years from manufacture.